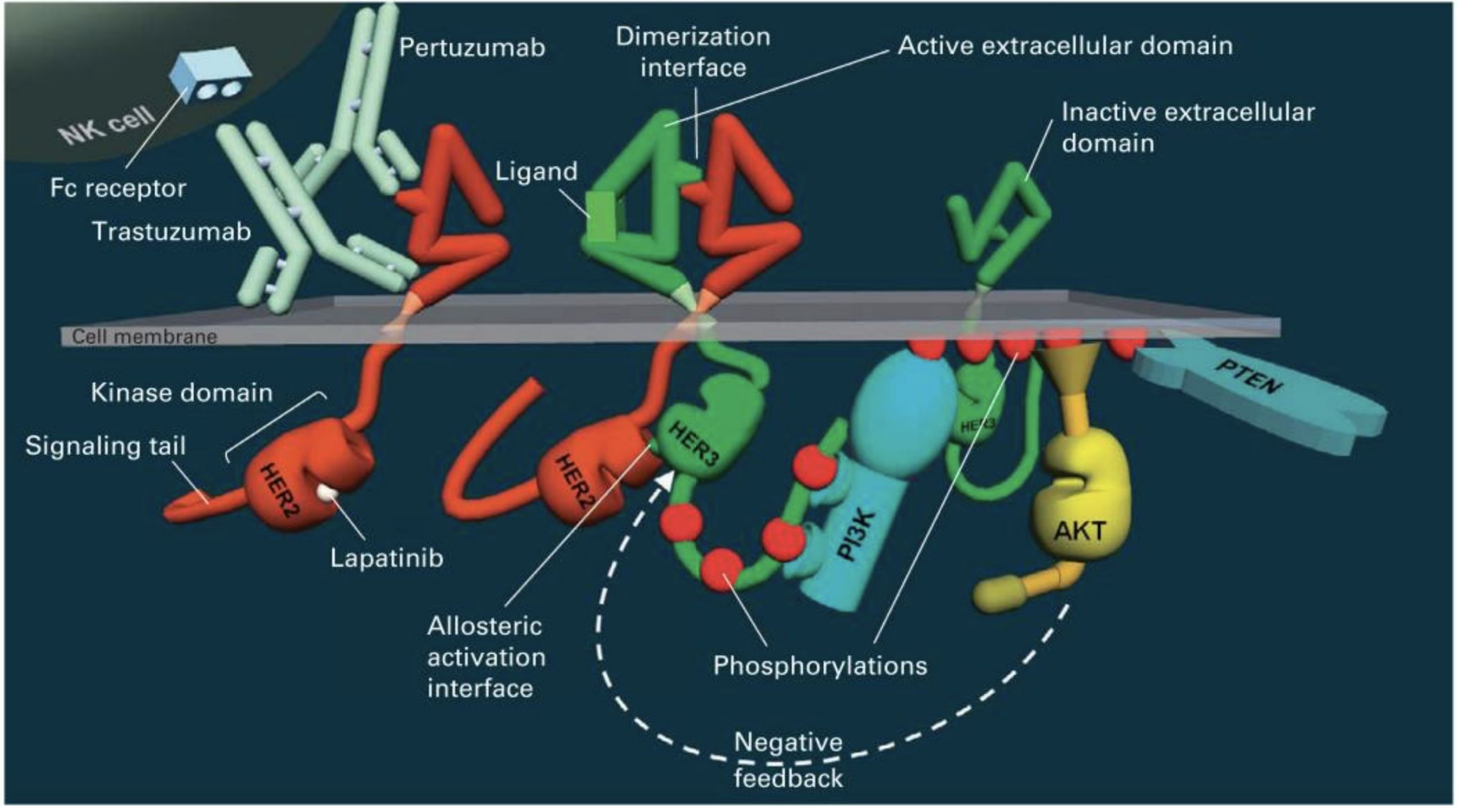
Moasser Lab

UCSF - Mission bay

Big picture

- *In HER2 amplified cancers, the tumorigenesis driver is the heterodimerization between HER2 and HER3*
- Prove that drugs should be targeting the **HER2** and **HER3** interaction below the cell surface as opposed to the **ECD**
- This was studied by using **Heregulin stimulation** and observing which pathways were activated
- Proved that even without HER3, pathways were still activated signifying HER2 may be able to self phosphorylate and activate these pathways


NK cell
Fc receptor
Trastuzumab
Pertuzumab
Dimerization interface
Active extracellular domain
Inactive extracellular domain
Ligand
Cell membrane
Kinase domain
Signaling tail
HER2
Lapatinib
HER2
HER3
Allosteric activation interface
PI3K
HER3
AKT
PTEN
Phosphorylations
Negative feedback

My projects

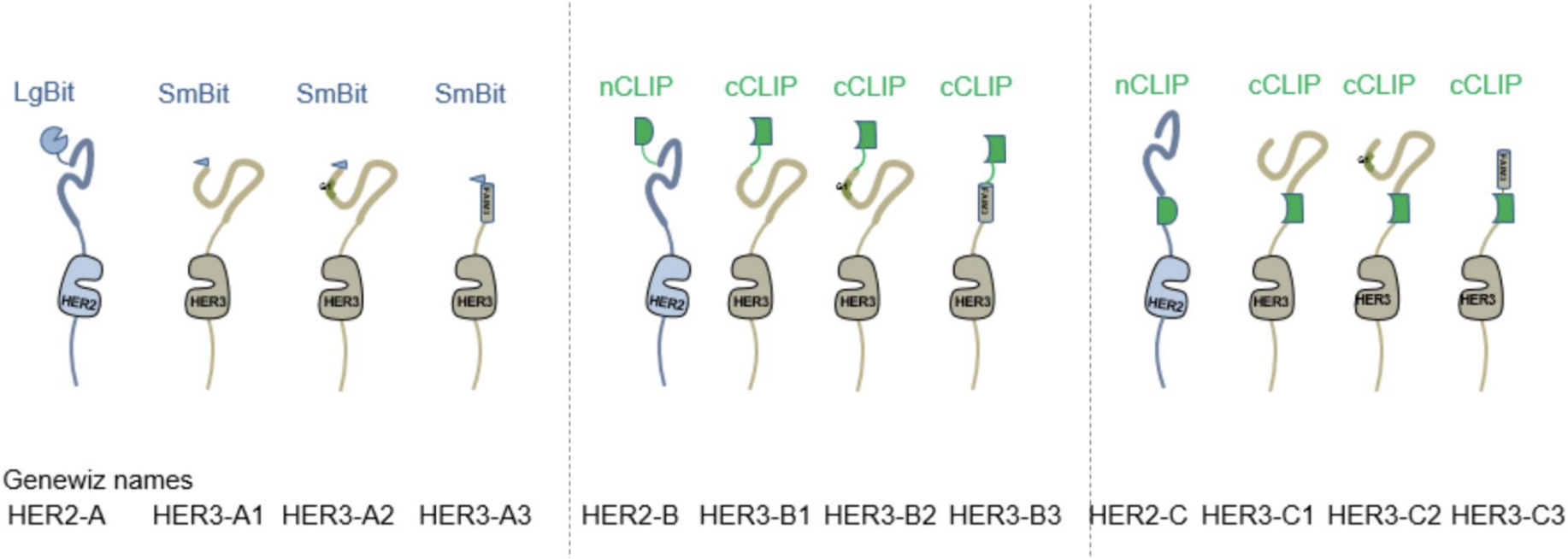
- Heregulin stimulation project
  - Western blot for paper
  - Studying HCC1569 cell lines with Her3KO-**Her3wt,** Her3KO-**Her3 Tailless**, Her3KO-**Luciferase** to see if *HRG stimulation still induces signaling*
  - Pathways incl. AKT, MAPK

- Light microscopy
  - CHO cells
  - Fixed and live cell
  - Use **Hoechst 33342** for live cell imaging
  - For fixed cell imaging, use **4% paraformaldehyde**

Techniques I use

- Western Blot
  - Cell culture
  - HRG stim.
  - Protein Purification
  - BCA
- Light microscopy
  - Cell culture
  - Transfection
  - HRG stim.
- Apoptosis Assay
  - FACS (Aria 3, 2-dye apoptosis assay)

Constructs being used



Other duties

Lab Manager duties:

- Drug, steroid, and antibacterial aliquots
- Solution and buffer inventory
    - TBST, LB, Transfer buffer, TAE
- Shipping and receiving

General duties

- Gateway cloning
- Bacteria transformation
- DNA extraction
- Mycoplasma testing
- Bacterial stocks
- Cell stocks

Previous positions & relevant techniques

- LBNL
  - *X-ray crystallography* - statistical analysis of calculated structure factors using **R**
- Conboy Lab
  - RT-PCR
  - Hybridoma maintenance
  - Tissue culture
- Courses
  - Hybridoma synthesis and isolation
  - ELISA and Flow

What I can add

- Hands on *wet lab experience* across disciplines: light microscopy, Western blot, FACS, Gateway cloning, Hybridoma technology, and experimental design.
- *Statistical analysis in R* of complex experimental datasets, including structure factor analysis for x-ray crystallography.
- Experience in different environments:
  - Lawrence Berkeley National Labs
  - Conboy Lab, Moasser Lab
- *Active local network* of **UCSF** and **UC Berkeley** associates facilitates collaboration.

Why this work is so interesting..

- *The cell atlas initiative has the potential to change the way we see and study molecular cell biology forever.* Being able to contribute to something of this scale is a big source for my passion for science.
- Access to **instruments** and **infrastructure** not traditionally available in academic settings
- **Nonprofit structure** means that passion for science drives the agenda, not the other way around
- Impetus for discovery is curiosity based